



Toward Dependable LLM-Driven Robot Software

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COLLAB · ROBOLAB



Two LLM roles, two dependability problems

FAULT PREVENTION

LLMs translate natural-language goals into actions on the robot's middleware

The model commands a running system.

RESEARCH QUESTION

Which factors have an impact on robot reliability?

FAULT REMOVAL

LLMs troubleshoot the software stack when it fails

The model reads a system that has already failed.

RESEARCH QUESTION

Can a model diagnose and repair faults?



01

FAULT PREVENTION

RQ 1

Which factors have an impact on robot reliability?



Empirical study on a simulated TIAGo

FAULT PREVENTION

COMMAND GRANULARITY →

LOW

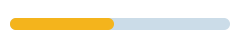
MID

HIGH

Validation



No validation



LOW `publish_twist, set_joint_positions`

MID `navigate_to, pick, place`

HIGH `fetch_object, deliver_to`

MODELS SO FAR

Qwen (3, 3.6, 3.8), Gemma 4, gpt-oss, GLM-4.5, Mistral Small 4

Which factors have an impact on robot reliability?

PRELIMINARY FINDINGS

- Success 43% (LOW), 47% (MID), 86% (HIGH)
- The validation layer halves executed errors (0.231 to 0.118) without hurting task success
- Success climbs with parameters: 8B 50%, 14B 56%, 32B 63%.



02

FAULT REMOVAL

RQ 2

Can a model diagnose and repair faults?



The eleven categories

FAULT REMOVAL

Process death / OOM

ROBUST · ISSTA '24

CPU starvation

ISSTA '24

Network faults

ROBUST · ISSTA '24

**DDS / QoS
misconfiguration**

ISSTA '24

Topic data corruption

ROBUST · ISSTA '24

**Parameter /
config error**

ROBUST · ISSTA '24

TF / URDF breakage

ROBUST · ISSTA '24

Clock skew

ISSTA '24

Resource exhaustion

ROBUST · ISSTA '24

**Node-lifecycle
orchestration**

ROBUST · ISSTA '24

Hang / deadlock

ROBUST · ISSTA '24

+ no fault

benign

The model reads failures, names the fault.



Train on injected faults, test on real faults

FAULT REMOVAL

— KNOWN FAULT TYPES

Template-grouped cross-validation

Can it learn a known catalogue?
Folds grouped by scenario template, never by episode.

— UNSEEN FAULT TYPES

Leave-one-fault-type-out

Entire categories excluded from training: generalisation to faults never seen.

— REAL FAULTS

Runtime-reproducible faults from ROBUST

Human-reported, never injected; pre-registered contamination protocol, same leakage canary.

THE QUESTION ANSWERED BY THE SPLIT

Can injected faults substitute for real ones?

If injected training transfers to real faults, the method scales without annotation. A negative result is also informative.

TO BE RELEASED the injector, the dataset and the benchmark for ROS 2 fault diagnosis



Open questions for the audience

01

FAULT PREVENTION

Given that VLAs exist, is LLM control through tools still worth making reliable?

02

FAULT REMOVAL

We score diagnosis against a single root cause per failure. Is that the right target?



RECAP

01 Fault prevention

Catch bad commands before they execute. A study of LLM control through tools on a simulated TIAGo.

RQ 1

Which factors have an impact on robot reliability?

QUESTION FOR THE AUDIENCE

Given that VLAs exist, is LLM control through tools still worth making reliable?

02 Fault removal

Diagnose and fix faults after they manifest. A model fine-tuned on injected faults, to diagnose and repair ROS 2.

RQ 2

Can a model diagnose and repair faults?

QUESTION FOR THE AUDIENCE

We score diagnosis against a single root cause per failure. Is that the right target?



B

Backup

Fault prevention: the seven checks and the nine error classes. Fault removal: how the eleven were chosen, and the table.



The validation layer, check by check

FAULT PREVENTION

CHECK	WHAT IT READS	WHAT IT REFUSES
Schema	the tool's declared JSON schema	a call with missing or mistyped arguments
Range	physical limits from PAL's URDFs	an argument outside joint or velocity limits
Frame	coordinate frames in the TF tree	a pose expressed in a frame that does not exist
Precondition	declared preconditions and state	an action whose preconditions do not hold
Workspace	the collision-free workspace	a target that would collide or is unreachable
Rate limit	motion calls per turn window	a burst of motion commands
Unit sanity	argument magnitudes	degrees where radians are expected

Server-side schema validation stays on in both arms, as the MCP SDK does by default; the client-side layer is the experimental factor.



The nine error classes

FAULT PREVENTION

CLASS	WHAT IT MEANS	INTERCEPTED BY
Hallucinated tool	a tool that does not exist	tool existence check
Malformed arguments	wrong shape or type	schema check
Wrong units	degrees for radians, mm for metres	unit sanity, by magnitude
Wrong frame	a pose in the wrong reference frame	frame check, for nonexistent frames only
Bad sequencing	an action before its preconditions	precondition check
Unsafe command	outside limits, workspace or rate	range, workspace and rate-limit checks
Wrong tool	a valid tool for the wrong purpose	not interceptable; adjudicated
Premature success	claiming completion before it holds	episode-level; scored from the final message
Timeout mishandling	a stalled action left unhandled	episode-level

Six of nine are interceptable before execution; executed-error rate is compared across arms, per-class rates within an arm.



How the eleven categories were chosen

FAULT REMOVAL

Two faults are different categories only if the right diagnosis leads to a different fix.

SPLIT

ISSTA files network faults and DDS/QoS together under Messages. The fixes differ: repair the link, or the QoS contract on a healthy link. So they are two categories.

MERGED

ISSTA's 50 sub-categories become 11. Wrong parameter type and wrong value share mechanism and fix, so they are one category; the detail lives in the scenario.

ADDED

Neither source has it: they are bug corpora, this is an evaluation instrument. No fault makes false alarms measurable; it is the answer when nothing matches.

Grounded, then frozen. Every category maps to an ISSTA class and to ROBUST bugs with verified fixes; the three with no ROBUST example are marked as gaps. The taxonomy was frozen before any data was generated, so categories cannot be redrawn to fit the results.



The eleven, category by category

FAULT REMOVAL

CATEGORY	ISSTA CLASS	INJECTED BY	REMEDICATION
Process death / OOM	Nodes	signal, cgroup OOM	restart the process
CPU starvation	Timeliness	cgroup quota, stress	adjust resource limit
Network faults	Messages	tc / netem	fix the network
DDS / QoS misconfiguration	Messages, Names	ros2 param, sidecar	fix the QoS
Topic data corruption	Semantic Types	in-graph sidecar	fix the config file
Parameter / config error	Parameters	ros2 param	fix the parameter
TF / URDF breakage	TF, Calibration	ros2 param, signal	fix the transform
Clock skew	Timeliness	clock sidecar	resync the clock
Resource exhaustion	Timeliness, Contextual	cgroup, process	free the resource
Node-lifecycle orchestration	Launch, Names, Nodes	ros2 lifecycle, signal	restart lifecycle node
Hang / deadlock	Timeliness, Nodes	SIGSTOP	restart the process

One mechanism and one remediation per category; grounded in ROBUST and ISSTA 2024.

